

CPE 470/670 Fall 2025 Project Checkpoint

Instructor: Parikshit Maini

Submission Deadline: 11:59 pm Thursday, November 20, 2025

Demo and Oral Evaluation - **Venue: SEM 342**; Time: 10 am - 12 noon Friday, November 21, 2025

All text answers must be written using Latex. Handwritten or scanned submissions will not be accepted. All code must be written in Python. Zip together all submission material and submit the Zip folder. All questions are compulsory.

This is a project deliverable and must be completed and submitted as a team. Each team member must be fully conversant with all parts of the deliverable. In addition to the submission, each team member will be individually evaluated in an oral exam at the time of the demo. You are not permitted to seek help from people outside the course or AI based tools. You are permitted to refer to class notes, lecture materials, books and other online teaching resources. You are not permitted to use existing implementations or library functions except for standard basic functionality unless explicitly approved by the teaching staff. In case of doubt check with the Instructor or TAs.

Consider a rectangular enclosed environment as shown in the figure below (figure shows the top view) with a ArUco marker printed on one of the walls. This environment has been created in room SEM 342 (access code 5791). The origin of the frame of reference along with two axes is marked on the floor.

Mount the camera sensor on your robot such that the camera center is aligned with the robot center and camera's optical axis is aligned with the front direction of the robot. Similarly mount the lidar on your robot.

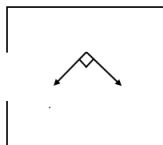


Figure 1: Top view of an instance of the operating environment for deliverable 2

Question 1a (10 points) Mount the LiDAR sensor on the robot and connect to RPi. Identify the geometric center of your robot and measure the mounting location from the center of the robot so you know its relative pose from robot center. You can use industrial velcro available with the TAs during office hours to mount the lidar. Write ROS code in Python to interpret the LiDAR data packet and retrieve the point cloud in polar coordinates.

Question 1b (10 points) Place the robot at the marked location in the environment in SEM 342, collect LiDAR data of the environment and use the data to detect the opening in the environment. You need to detect and print the (x,y) coordinates of both points of the opening. At the time of evaluation, the environment, including opening, origin location and orientation of the axes may be modified to verify your code. The reference frame is fixed to be orthogonal. You can use any library function calls.

Question 1c (20 points) Explain your solutions for questions 1a and 1b in a text document. Readability and presentation of your solutions in the text document will also be considered in the grade.

Question 2 : An ArUco marker is printed and mounted on the wall of the environment. The ArUco marker is selected from the 4X4-50 dictionary. Estimate the pose of the ArUco marker using OpenCV library functions. To ensure high accuracy in obtaining the marker pose, it is crucial to have a well-calibrated camera. Use camera calibration matrix from lab 6 for your camera. You can use the function `cv2.aruco.estimatePoseSingleMarkers()` with the computed calibration parameters to get the marker pose. The marker pose returns two 3-dimensional vectors, the translation vector, `t_vec` and the rotation vector, `r_vec`. The two vectors provide a 3d transformation from the marker coordinate system to the camera coordinate system.

We define **alignment error**, α , to be the angle between the robot orientation and straight line connecting the robot to the center of the ArUco marker. Since the camera center is aligned with the robot center and the optical axis of the camera is aligned with the robot's positive x-axis (pointing out of the nose of the robot), horizontal pixel distance between the marker center in the camera image and center of the image frame can be used to compute alignment error. Since horizontal field of view of the camera is 60° , you can use the ratio of pixel distance between marker center and center of the image frame to compute α .

$$\alpha = \frac{(\text{horizontal pixel distance of marker center from center of image frame})}{\text{width of the image frame in pixels}} \cdot 60$$

We define **distance separation**, δ , to be the euclidean distance between the robot center and the ArUco marker. The translation vector, `tvec`, is centered at the ArUco marker's center OpenCV Documentation. Use `t_vec` returned by the ArUco Tag pose estimation to compute δ . Note that the dimension perpendicular to the ground plane, will not contribute to δ .

Question 2a (20 points) : Write a ROS node that computes the alignment error, α , and distance separation, δ , of the **robot** w.r.t. the ArUco marker as defined in the above text. Publish α and δ as Float values in a single custom ROS message.

Question 2b (20 points) : Subscribe to the custom topic that publishes α and δ . Use α to adjust the robot's orientation angle with a constant angular velocity of 0.005 rad/s until the alignment error α is less than 3° . Use `/cmd_vel` topic to publish the angular velocity to center the ArUco in your image frame. You must use the alignment error messages as feedback for aligning your robot in the direction of the ArUco marker. Once the robot is aligned with the ArUco, print the alignment error. Also, print current value of δ .

Question 2c (20 points) : Now that the robot is aligned in the direction of the ArUco marker, move it 30 centimeters towards ArUco marker. This can be achieved by utilizing the wheel encoders on the robot to measure how far the wheels have moved over time. You must use these encoders as feedback to determine how far the robot has traveled. Publish a constant linear velocity of 0.20m/s to the `/cmd_vel` topic to move the robot until it has traveled your desired distance. Finally, print the distance the robot has traveled and the current value of δ .

Question 2d (20 points) Explain your solutions and computations for questions 2a, 2b and 2c in a text document. Readability and presentation of your solutions in the text document will also be considered in the grade.

Question 3: Evaluate your team's contributions in deliverable 2 as follows. For a team of size N , total team contribution = $N \times 10$. Submit each person's contribution out of 10, including your own.